

PENG CHENG

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Research Interests: AIGC security, IoT security

PROFESSIONAL EXPERIENCE

Researcher, State Key Laboratory of Blockchain and Data Security, Zhejiang University

June 2024 – Present

Member of the AI Data Security Team

- Conducted research on AI-generated content (AIGC) security
 - Developing the *DFscan platform* for multimodal deepfake detection. Leading the technical team in the platform design and development
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Qiusi Research Fellow, ZJU-Hangzhou Global Scientific and Technological Innovation Center

December, 2023 - May, 2024

Member of the Cyberspace Security Research Institute.

- Conducted research on multimodal data security and privacy protection technologies.
 - Developed multimodal deepfake detection systems.
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Postdoctoral Researcher, Zhejiang University

January, 2021 - November, 2023

College of Computer Science and Technology

- Researched voice security and privacy protection in human-computer interaction scenarios.
 - Postdoctoral supervisor: Prof. Kui Ren (Qiusi Chair Professor, AAAS/ACM/CCF/IEEE Fellow, Dean of the College of Computer Science and Technology of Zhejiang University)
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EDUCATION

Ph.D. Degree, Lancaster University

October, 2016 - December, 2020

School of Computing and Communications

- Major: Computer Science
 - Thesis: *Acoustic-channel Attack and Defence Methods for Personal Voice Assistants*.
 - Supervisors: Prof. Utz Roedig (Full Professor of Computer Science at University College Cork) and Prof. Jeff Yan (Full Professor of Cyber Security at the University of Southampton)
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Dual Master's Degree, KU Leuven, Belgium & Tsinghua University, China

September, 2012 - September, 2015

Department of Electrical Engineering, KU Leuven & School of Integrated Circuits, Tsinghua University

- Major: Electrical Engineering (KU Leuven) & Integrated Circuit Engineering (Tsinghua University)
 - Supervisor: Prof. Guoqiang Bai (Associate Professor at Tsinghua University)
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B.Eng. Degree, Beijing University of Posts and Telecommunications, China

September, 2008 - June, 2012

School of Electronic Engineering

- Major: Electronic Science and Technology

RESEARCH PROJECTS

Total Funding: £7.01M (British Pounds; all figures in GBP)

National Natural Science Foundation of China (NSFC)

- Principal Investigator: “Research on Speech Synthesis Data Compliance Management Technology Based on Intrinsic Characteristics of Audio Signals”, (2025-2028, NSFC General Program Project, £54K)
- Participant: “High-Performance Visual Perception Models Using Deep Learning” (2023–2026, £59K)
- Participant: “Cross-Chain Security in Heterogeneous Blockchain Networks” (2023–2027, NSFC Key Project, £309K)
- Participant: “Research on Voice Attack and Defense Based on the Physical Characteristics of Smart Device Sensing Components” (2022–2025, £64K)

Key R&D Programs of China

- Participant: “Multimodal Network Environment Construction Technology Based on Public Cloud-Network Resources”, 2024-2027 (£1.64M)
- Participant: “Aggregation and Transfer of Machine Learning Models” under the *National Science and Technology Innovation 2030 Initiative - New Generation Artificial Intelligence*, 2021 – 2025 (£1.51M)
- Participant: “Security Protection Technology for Industrial Control Programming Platforms Based on Domestic Cryptographic Algorithms”, 2022 – 2024 (£1.60M)

Provincial, Municipal, and University-Level Research Projects

- Participant, the Key R&D Programme of Zhejiang Province, 2025 (£272K)
- Participant, Hangzhou Key R&D Program: “Key Technologies and Platform Development for Security Detection of Large AI Models”, 2024 – 2027 (£1.83M)
- Principal Investigator, Hangzhou West Innovation Corridor Development Special Fund: “Toolchain for Deep Synthetic Content Analysis Based on Physical Attribute Attribution”, 2024 – 2026 (£33K)

Industry Collaboration

- Principal Investigator, **Zhejiang University-Alibaba Collaboration Project**: “Active and Passive Security Protection Technologies for the *Maojing* Voice Interaction System”, 2025 – 2026 (£41K)
- Participant, **Zhejiang University-Ant Group Joint Laboratory Industry Collaboration Project**: “Security Risk Detection and Alignment Strategies for Large Model-based AI Agents”, 2025 (£65K)
- Participant, **China Southern Power Grid Research Institute Co., Ltd.**: “Research and Development of AI-Driven Automated Security Detection Technologies and Components for Power Systems”, 2024 – 2026 (£255K)
- Participant, **CRRC Zhuzhou Electric Locomotive Research Institute Co., Ltd.**: “Deep Learning Algorithm Evaluation and Security Verification Platform”, 2024 – 2027 (£127K)
- Participant, **China Southern Power Grid Research Institute Co., Ltd.**: “Research on AI Attack-Defense Library Design and Test Component Development for New Power System Scenarios (2023)”, 2023 – 2025 (£163K)

PUBLICATIONS

Five Most Relevant/Representative Publications:

- Ba, Z., Zhong, J., Lei, J., **Cheng, P.***, Wang, Q., Qin, Z., Wang, Z., Ren, K. *SurrogatePrompt: Bypassing the Safety Filter of Text-to-Image Models via Substitution*. Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS 2024), 1166–1180.
- **Cheng, P.**, Wang, Y., Huang, P., Ba, Z., Lin, X., Lin, F., Lu, L., Ren, K. *ALIF: Low-Cost Adversarial Audio Attacks on Black-Box Speech Platforms Using Linguistic Features*. IEEE Symposium on Security and Privacy (SP 2024), 1628–1645.
- Ba, Z., Wen, Q., **Cheng, P.***, Wang, Y., Lin, F., Lu, L., Liu, Z. *Transferring Audio Deepfake Detection Capability Across Languages*. Proceedings of the ACM Web Conference (WWW 2023), 2033–2044.

- **Cheng, P.**, Wu, Y., Hong, Y., Ba, Z., Lin, F., Lu, L., Ren, K. (2023). *UniAP: Protecting Speech Privacy With Non-Targeted Universal Adversarial Perturbations*. IEEE Transactions on Dependable and Secure Computing (TDSC, JCR Q1), 21(1), 31–46.
- **Cheng, P.**, Roedig, U. (2022). *Personal Voice Assistant Security and Privacy—A Survey*. Proceedings of the IEEE, 110(4), 476–507.

Other Publications (Reverse Chronological Order)

- Wen, Q., Li, H., Ba, Z., **Cheng, P.**^{*}, He, M., Lu, L., Ren, K. (2026). *HyperPotter: Spell the Charm of High-Order Interactions in Audio Deepfake Detection*. International Conference on Machine Learning (ICML 2026).
- Li, Q., Lin, Y., Lian, W., Ba, Z., **Cheng, P.**^{*}, Lian, Z. (2026). *MixFake: Benchmarking and Enhancing Audio Deepfake Detection in Diverse Real-world Mixed Audio*. IEEE International Conference on Multimedia and Expo (ICME 2026), **Spotlight**.
- Liu, Q., Zhang, Y., Ba, Z., Shuai, C., **Cheng, P.**, Zheng, T., Wang, Z. *Attack-Resistant Watermarking for {AIGC} Image Forensics via Diffusion-based Semantic Deflection*. To appear at the 14th International Conference on Learning Representations (ICLR 2026).
- Ba, Z., Yi, L., **Cheng, P.**^{*}, Li, Q., Wang, Q., Lu, L. *Beyond Content: A Comprehensive Speech Toxicity Dataset and Detection Framework Incorporating Paralinguistic Cues*. To appear at the 40th AAAI Conference on Artificial Intelligence (AAAI 2026).
- Dong, Z., Shuai, C., Ba, Z., **Cheng, P.**, Qin, Z., Wang, Q., Ren, K. (2025). *WMCopier: Forging Invisible Watermarks on Arbitrary Images*. The 39th Conference on Neural Information Processing Systems (NeurIPS 2025).
- REN, K., LIN, F., BA, Z., LIU, Z., **Cheng, P.** (2025) *Deepfake detection: key challenges and technical approaches*. Computing Magazine of the CCF. 1(2): 8–15.
- Huang, P., Pan, K., Wang, Q., **Cheng, P.**, Lu, L., Ba, Z., Ren, K. (2025). *SecHeadset: A Practical Privacy Protection System for Real-time Voice Communication*. Proceedings of the ACM MobiSys 2025.
- Ba, Z., Gong, B., Wang, Y., Liu, Y., **Cheng, P.**^{*}, Lin, F., Lu, L., Ren, K. (2024). *Indelible “Footprints” of Inaudible Command Injection*. IEEE Transactions on Information Forensics and Security (TIFS, JCR Q1).
- Huang, P., Wei, Y., **Cheng, P.**, Ba, Z., Lu, L., Lin, F., Wang, Y., Ren, K. (2024). *Phoneme-Based Proactive Anti-Eavesdropping with Controlled Recording Privilege*. IEEE Transactions on Dependable and Secure Computing (TDSC, JCR Q1).
- Huang, P., Wei, Y., **Cheng, P.**, Ba, Z., Lu, L., Lin, F., Zhang, F., Ren, K. *InfoMasker: Preventing Eavesdropping Using Phoneme-Based Noise*. Network and Distributed System Security Symposium (NDSS 2023).
- **Cheng, P.**, Sankar, M. S. A., Bagci, I. E., Roedig, U. *Adversarial Command Detection Using Parallel Speech Recognition Systems*. Computer Security - ESORICS 2021 International Workshops, 238–255.
- **Cheng, P.**, Bagci, I. E., Roedig, U., Yan, J. (2020). *SonarSnoop: Active Acoustic Side-Channel Attacks*. International Journal of Information Security (IJIS, JCR Q2), 19(2), 213–228. **Finalist for the “Most Innovative Research” Pwnie Award at Black Hat USA 2019.**
- **Cheng, P.**, Bagci, I. E., Yan, J., Roedig, U. *Smart Speaker Privacy Control—Acoustic Tagging for Personal Voice Assistants*. IEEE Security and Privacy Workshops (SPW 2019), 144–149.
- **Cheng, P.**, Bagci, I. E., Yan, J., Roedig, U. *Towards Reactive Acoustic Jamming for Personal Voice Assistants*. Proceedings of the 2nd International Workshop on Multimedia Privacy and Security (2018), 1–13.

Preprints/Under Review

- Gong, B., **Cheng, P.**^{*}, Ba, Z., Lu, L., Tang, J., Hou, J., Xue, S., Ren, K. (2025) *Bridging the Synthesis-Detection Gap: A Chinese Audio Deepfake Dataset and Industrial-Scale Retrieval-Augmented Detection*.
- Ba, Z., Zhang, Y., **Cheng, P.**^{*}, Gong, B., Zhang, X., Wang, Q., Ren, K. (2025). *Robust Watermarks Leak: Channel-Aware Feature Extraction Enables Adversarial Watermark Manipulation*. arXiv:2502.06418
- Gong, B., Shuai, C., Wen, Q., **Cheng, P.**, Wang, Q., Ba, Z., Wang, Z., Ren, K. (2025). *Test-Time Adaptation for Audio Deepfake Detection*.
- Ba, Z., Fu, H., Yang, Y., Chen, H., Wang, Q., **Cheng, P.**, Qin, Z., Ren, K. (2025). *JudgeRail: Harnessing Open-Source LLMs for Fast Harmful Text Detection with Judicial Prompting and Logit Rectification*.
- Wen, Q., **Cheng, P.**, Ba, Z., Yi, L., Qin, Z., Lu, L., Wang, Q., Ren, K. (2024). *CLINDA: A Cross-lingual Domain Adaptation Framework for Challenging Audio Deepfake Detection Tasks across Languages*.
- Lei, J., Wang, Q., **Cheng, P.**, Ba, Z., Qin, Z., Wang, Z., Liu, Z., Ren, K. (2023). *Masked Diffusion Models Are Fast and Privacy-Aware Learners*. arXiv:2306.11363

Thesis

- **Cheng, P.** (2020). *Acoustic-Channel Attack and Defence Methods for Personal Voice Assistants*. Ph.D. Dissertation, Lancaster University. Granted Patents

STUDENT SUPERVISION AND MENTORSHIP

- Co-supervising **5 Ph.D. students**, **7 Master's students** and **one undergraduate student** at Zhejiang University on research projects in AIGC security and multimodal privacy
- Co-supervised a Zhejiang University Master's student to win the **National Graduate Scholarship (China)-the highest-level scholarship awarded to Master's students in China** (Oct 2024).
- Successfully mentored **1 PhD student**, **5 Master's students**, and **5 undergraduates** to degree completion, guiding thesis design, research execution, and publication strategies.
- Teaching Assistant for lab/practical sessions: *CS4615: Computer Systems Security*, University College Cork, Ireland; *SCC110: Software Development*, Lancaster University, UK.

HONORS & AWARDS

- **Third Place** | *IJCAI 2025 Deepfake Detection Challenge* (2025), **Advisor**, Track: Audio-Visual Detection and Localization (DDL-AV)
- **National Grand Prize (Top-Tier Award)** | *19th "Challenge Cup" National Competition for Extracurricular Academic Science and Technology Works* (2024), **Advisor** (Ranked 2nd among 3 advisors), Project: "*Multimodal AI Audit Matrix: Deepfake Detection and NSFW Content Regulation Platform*"
- **Top 5 Nationwide** | *3rd China Artificial Intelligence Competition* (2021), **Primary Advisor** (Ranked 1st among 2 advisors), Track: *Audio Deepfake Detection Under Open-Speaker Scenarios*.
- **Top 5 Nationwide** | *3rd China Artificial Intelligence Competition* (2021), **Primary Advisor** (Ranked 1st among 2 advisors), Track: *Speaker-Specific Audio Deepfake Detection*.
- **Finalist for "Most Innovative Research" Pwnie Award** | *Black Hat USA 2019*, **First Author** (1st of 4 contributors) for research titled "*SonarSnoop: Active Acoustic Side-Channel Attacks*".
- **Postdoctoral Excellence Grant (Second Class)** | Zhejiang Provincial Department of Human Resources and Social Security (Aug 2021).
- **Ph.D. Scholarship** | Faculty of Science and Technology, Lancaster University, UK (2016–2020).

IMPACTS

Industry Contributions

1. Vulnerabilities in Text-to-Image AI Models

- Proposed *Surrogate Prompt*, a prompt attack method to bypass safeguards of commercial large-scale AI models (e.g., Midjourney, Stability.ai), enabling systematic generation of policy-violating content. Findings were **acknowledged and taken into consideration for security improvement** by these leading vendors.

2. Attacks on AIGC Watermarking Systems

- Developed *WMCopier*, an adversarial watermark forgery method leveraging diffusion inversion techniques to compromise commercial watermarking systems. **Recognized by Amazon's Responsible AI Team for identifying critical vulnerabilities**, leading to collaborative improvements in their defensive frameworks. Received official thank-you letter.

3. Open-Source Contributions

- Developed and open-sourced *ALIF*, the first black-box adversarial attack framework leveraging linguistic features to generate **imperceptible adversarial audio** that effectively bypasses human perception while compromising speech AI systems. **Adopted by NVIDIA** for integration into their **official AI security toolkit** to advance red teaming capabilities (*under active development*).

Media Recognition

- **Developed** *SonarSnoop: Active Acoustic Side-Channel Attacks*, a novel methodology exploiting smart device speakers and microphones to extract sensitive data. Garnered attention from IT media (*Motherboard*, *ZDNet*, *Sophos*). Praised by **Bruce Schneier** (renowned cryptographer) and **Prof. Ross Anderson** (Well-known security researcher) on X for its novelty.

GRANTED PATENTS

- Qian, Y., Zhang, X., Wang, Q., Ba, Z., **Cheng, P.**, et al. “Review System, Method, Computer Device, and Medium for Image Sensitive Elements”, *China, Patent No. CN106610969A[P]*, Granted: 02-Apr-2025.
- Ba Zhongjie, Wu, Y., **Cheng, P.**, et al. “Privacy Protection Method and Device Using White-Box Speech Adversarial Examples”, *China, Patent No. CN2022109965917[P]*, Granted: 03-Dec-2024.
- Ba, Z., Zheng, Q., **Cheng, P.**, et al. “Enhanced Deepfake Image Detection Method and Device Based on Generative Adversarial Networks (GANs)”, *China, Patent No. CN2024100814592[P]*, Granted: 21-Jun-2024.
- Ba, Z., Wang, Y., **Cheng, P.**, et al. “Security Evaluation Method for Speech Recognition Models via Semantic Space Perturbation”, *China, Patent No. CN116758899B*, Granted: 13-Oct-2023.
- Huang, P., Ba, Z., **Cheng, P.**, et al. “Privacy Protection Method, System, and Medium for Voice Communication Based on Voice Obfuscation”, *China, Patent No. CN119449493B*, Granted: 08-Jul-2025.
- Gong, B., Huang, P., Ba, Z., **Cheng, P.**, et al. “Cross-Domain Detection Method and Device for Deep Synthetic Audio Based on Self-Supervised Auxiliary Tasks”, *China, Patent No. CN119479611B*, Granted: 29-Apr-2025.

ACADEMIC SERVICES

Journal Editorial Roles

- **Special Issue Initiator and Guest Editor-in-Chief:** Special Issue on “Intelligent Voice Security and Defense Technologies”, *Journal of Cyber Security*, 2025.

Conference Program Committees

- **Conference Reviewer:** ICML 2026, ICLR 2026, ACM MM 2026, IEEE SLT 2026, AAAI 2026, ACM Web Conference (WWW 2025).

Journal Reviewer Roles

- **English Journals:** *Proceedings of the IEEE*, *IEEE Transactions on Information Forensics and Security (TIFS)*, *IEEE Transactions on Dependable and Secure Computing (TDSC)*, *IEEE Internet of Things Journal (IoT-J)*, *ACM Transactions on Internet of Things (TIOT)*.
- **Chinese Journal:** *Journal of Information Network Security*.

Reviewer Recognition

- **ICML 2026 Silver Reviewer** — top-tier recognition awarded by the ICML 2026 Program Chairs.

OTHER ACADEMIC ACTIVITIES

- Contributed to the preparation of a €720,839 grant proposal (*Science Foundation Ireland – SFI*) on *Security and Privacy of Personal Voice Assistants* under Prof. Utz Roedig (2019). The proposal is anchored in my doctoral research.
- Visiting Scholar at the Department of Computer Science, University College Cork (UCC), Ireland (2019–2020).
- Participated in the *São Paulo Advanced Science School (ESPCA)* on Smart Cities, hosted by the University of São Paulo (2017).
- Selected as **one of 75 global top graduate students and postdoctoral researchers** (sponsored by the *São Paulo Research Foundation – FAPESP*).

GOOGLE SCHOLAR PAGE

<https://scholar.google.com/citations?user=h3kfCEAAAAJ&hl=en>